Additional file 1: supplementary methods and sensitivity analyses

S1 Pretraining pools and data provenance

Single-chain inputs total 194887 gamma and 1262692 delta sequences before pretraining filters. All chains appearing in the paired corpus are excluded from pretraining, and sequences shorter than six residues are removed. The main pretraining training pool contains 187092 gamma and 1230589 delta sequences, with validation sets of 3818 and 25114 sequences. Training and validation are fully disjoint at the unique CDR3-sequence level. Source experiments use an additional study-excluded pool containing 186949 gamma and 1230167 delta training sequences, with 3815 and 25106 validation sequences. Both source protocols use the same filtered pretraining pool. SHA256 manifests identify raw, processed, split and provenance artifacts; training seeds, selected-checkpoint identifiers and actual exposure appear in Additional file 3.

S2 Corruption and reconstruction

Each single-chain sequence is assigned a masking rate of 15% or 30% with equal probability. Exactly the larger of one and the rounded rate times sequence length is selected without replacement. Each selected residue independently receives a mask token (80%), a uniform canonical amino acid (10%) or its original token (10%). Deletion starts are sampled uniformly, with span lengths uniformly 1–3, until at least the rounded 8% target is covered; spans can overshoot that target. Deletion follows masking and can overlap selected sites. The damaged mask includes selected-but-unchanged residues and deleted sites. The aligned-copy control copies the corrupted aligned token and uses a mask at deleted sites. Reconstruction includes EOS in token NLL; residue accuracy is separately evaluated on original residue positions. No reconstruction metric is interpreted as partner recovery.

S2.1 Denoising pretraining improves single-chain reconstruction

On the frozen square-root-pretrained validation corruptions, selected-residue accuracy is 71.3% and 70.3% for gamma and 64.1% and 64.7% for delta across the two seeds. Corresponding aligned-copy accuracy is 6.9%/6.3% for gamma and 6.7%/6.2% for delta. This shows reconstruction beyond the specified aligned-copy sanity control. It does not establish that pretraining improves recorded partner recovery; that question requires the downstream matched comparisons.

S3 Generator training and matched controls

The generator is a shared encoder–decoder Transformer with 7396890 parameters, four encoder and four decoder layers, width 256, eight attention heads, feedforward width 1024 and dropout 0.1. It tokenizes individual amino acids with chain identity and control tokens, using a maximum of 64 tokens. The two conditionals share parameters:

$$p_\theta(d \mid g), \qquad p_\theta(g \mid d). \tag{1}$$

Training minimizes teacher-forced target-token negative log likelihood (NLL), including the end-of-sequence (EOS) token and excluding the chain-control prediction from the loss.

Denoising uses 20000 steps, batch size 1024, AdamW at learning rate 3×10^{-4} and 500 warmup steps. Each sequence is masked at 15% or 30% with equal probability. Selected residues are replaced by mask/random/unchanged tokens at 80/10/10 proportions, followed by short-span deletions targeting approximately 8% of positions. Reconstruction is evaluated separately on corruption-selected residues and against an aligned copy control. The saved "damaged" mask includes selected-but-unchanged positions in the 80/10/10 rule as well as deleted positions; it is not restricted to visibly altered residues. These teacher-forced metrics are not free partner-generation accuracy. Raw-frequency, square-root, balanced, gamma-only and delta-only chain mixtures use the same total number of sequence draws. Sequence and target-token exposure are reported separately. The square-root checkpoints are reused from the pair-excluded historical pretraining, with independently evaluated frozen corruption.

Paired fine-tuning uses at most 1500 steps at batch size 256, learning rate 2×10^{-5}, 200 warmup steps, validation every 100 steps and patience five. The selection criterion averages known-partner NLL within query, then queries within direction. Bidirectional models average the two directions equally; directional exposure diagnostics select using their specified direction only. Sampling chooses modeled directions equally, input queries uniformly and one recorded partner uniformly per query. The constant-source control changes only source residues while retaining initialization, target sampling and optimization. Random initialization changes only initial weights. It matches downstream training settings but receives no denoising pretraining, so this is not a matched-total-computation comparison. Shuffled-input evaluation applies a deterministic query permutation to the fitted conditional model, disrupting the input–recorded-partner association. Edge-uniform sampling is a separate control. Table S1 summarizes the roles of these four controls.

The unidirectional sensitivity controls receive at most 750 steps and 100 warmup steps, matching nominal per-direction sequence exposure. Directional exposure diagnostics instead retain the 1500-step maximum. Actual exposure under early stopping is recorded. Data-efficiency comparisons use nested complete-component subsets and both pretrained and random models. The 10/25/50/100% labels correspond to 725/1814/2249/7254 edges. The nominal 50% subset is only 31.0% because the 5005-edge component cannot be split. The curve therefore changes component composition as well as sample size. All three smaller subsets consist exclusively of singleton components.

The conditional and constant-source generators share initialization, target sampling, optimizer and nominal downstream budget. Constant-source input keeps the chain token and removes

source residues. Random initialization removes pretraining and retains downstream settings; total training computation is not matched. Shuffled-input evaluation permutes queries without fixed points and keeps the original recorded-partner labels. A separate copy of the fine-tuned generator initializes the ranker. Its copied encoder is trainable; the source generator is not updated. All paired fits use the same protocol-specific training edges. Seeds, selected checkpoints and actual exposure accompany Additional file 3.

AdamW uses weight decay 0.01 and gradient-norm clipping at 1.0. For denoising and generator training, the per-step learning rate is the nominal rate multiplied by $\min(1, s/w)$ and $[1+\cos(\pi \min(1, s/S))]/2$, where s is the current step, w the warmup and S the nominal maximum steps. This applies cosine decay over the whole nominal run, including warmup. Ranker training uses its fixed stated learning rate. Training may stop early; actual exposures, not only nominal limits, are retained.

Table S1: Core generator controls. All conditions retain the protocol-specific test query population and fixed primary sampling rule.

Condition	Comparison purpose
Pretrained conditional	Uses the input chain after denoising pretraining.
Constant-source target prior	Removes source residues while matching initialization, target sampling and downstream optimization.
Random initialization	Removes denoising pretraining; the downstream budget is matched, total computation is not.
Shuffled input	Breaks the input–recorded-partner association at evaluation using the fitted conditional model.

S4 Secondary ranker and reference scorers

The pair ranker evaluates whether an additional scoring stage helps order the generator's candidates; it is a secondary evaluation component. The ranker starts from a separate copy of the selected condition-specific generator's encoder, after paired fine-tuning. It uses amino-acid-only pooling, 128-dimensional chain-specific projections, a normalized vector dot product and a two-layer interaction head with hidden width 256 and dropout 0.15 over the concatenated vectors, their product and absolute difference. The copied encoder and ranking head are optimized on the same training edges used by the generator. The original generator remains frozen, but the copied encoder is trainable. Separate training refers to the subsequent parameters and optimization; neither the representations nor the cohort are independent. Training uses 3000 steps at batch size 256 and learning rate 3×10^{-4}. The first 1500 steps use multi-positive contrastive loss; the remaining steps use constructed-negative binary cross-entropy (BCE). The primary negative ratio is three per positive, with half random and half length/terminal-motif matched draws. Ratios one and five and random-only training provide sensitivity controls. All negative construction uses training edges only; known training partners are excluded. The best mean directional selection-validation MRR after explicit-negative training begins selects the checkpoint. Positive and constructed-negative classes receive equal BCE loss weight at every ratio; increasing the ratio changes negative sampling diversity rather than the total weight of the negative class.

Fixed-pool scorers include conditional likelihood, target-prior likelihood, deterministically shuffled-input likelihood, their conditional-minus-prior likelihood difference, ranker, fused score and seeded random scores. Likelihood scores are negative per-target-token NLL, including EOS; thus the likelihood difference uses length-normalized log scores rather than the raw sequence log-probability ratio. Additional controls are training-target frequency, a shallow length/terminal-feature logistic model and an exact two-chain edit-neighbor score. For a candidate (g, d) and training edge set E_{tr}, the latter is

$$s_{\mathrm{NN}}(g, d) = - \min_{(g', d') \in E_{\mathrm{tr}}} \frac{1}{2} \left[\frac{\mathrm{Lev}(g, g')}{\max(|g|, |g'|)} + \frac{\mathrm{Lev}(d, d')}{\max(|d|, |d'|)} \right]. \tag{2}$$

Every training edge contributes to the minimum. The shallow control is fitted to training positives and mixed constructed negatives at ratio 1:3, selecting $C \in \{0.1, 1, 10\}$ by mean Hit@5 across the four small selection-validation pools.

A complementary nearest-input partner-transfer baseline finds the closest training input using normalized full-CDR3 Levenshtein distance, resolves ties lexicographically, and proposes that input's recorded training partners. Targets are ordered by descending training degree and then lexicographically, retaining at most 64 unique candidates. A fixed five-neighbor sensitivity concatenates those lists in neighbor order. No test target is injected into these proposals; under exact target exclusion their recovery is necessarily zero. For frozen retrieval pools, a separate nearest-input partner-similarity score ranks each candidate by its minimum normalized distance to any partner of the single nearest training input. This similarity extension scores truth-containing pools and is not direct partner transfer.

S5 Candidate construction and score calibration

The primary ranker first uses multi-positive contrastive training, then equally weighted positive and constructed-negative binary cross-entropy. Three negative draws per positive are drawn from supervised-training combinations, excluding known training partners. The mixed policy combines random and length/terminal matching; actual fallbacks and draw counts are recorded. Unrecorded combinations are not biological negatives. Sensitivities change ratios to one or five, or use random-only draws. Fixed retrieval pools inject all role-specific recorded positives and up to 100 unrecorded targets from that role's universe. Other recorded partners are excluded from distractors. Random, length, terminal-motif and normalized-edit matching are deterministic at fixed seed. All scorers use the same frozen lists. Free sampling instead makes 64 draws with temperature one and top-p 0.95; 6–35 canonical residues and observed EOS are required. Invalid and duplicate draws are never replaced. Training-target exclusion is an explicitly separate policy that can delete legitimate exposed positives.

For each negative draw, the source direction is chosen with equal probability. Under the mixed policy, a hard draw is chosen with probability 0.5. Candidate training targets within one residue of the reference target length are sorted by absolute length difference, mismatch count over the first and last three residues, then sequence. The draw is uniform over the first 64 eligible targets after known-training-partner exclusion. If none remain, sampling falls back to uniform random eligible training targets. Sampling is with replacement; no held-out pairing labels filter training

negatives. The two BCE classes each receive one-half of the loss weight at every negative ratio.

Fixed-pool length matching sorts eligible role targets by the minimum length gap to a role-specific positive, then normalized full-CDR3 edit distance and sequence. Motif matching uses the sum of Levenshtein distances over the first and last three residues, then full normalized edit distance and sequence. Edit matching uses full normalized distance and sequence. Random pools sample without replacement. Candidate targets are lexicographically stored, and score ties use that order. Truth-containing retrieval is an explicitly constructed diagnostic; no such positive-directed pool construction is used in free sampling.

The primary setting uses 64 draws at temperature one and top-$p = 0.95$. Split-17 primary conditional and constant-source models save 256 draws and reuse prefixes of 16, 64 and 256. A prespecified six-cell temperature/top-p grid is inspected on split-17 tuning data but does not replace the primary setting. Validity requires 6–35 canonical residues and an observed EOS token, allowing 36 decoding steps. Invalid or duplicate draws are not replaced. Candidates are evaluated both with and without exact training targets. The exclusion policy may remove legitimate recorded positives in exposed-target settings, and is reported only as a separate sensitivity analysis. Free generation does not guarantee that any recorded partner exists among candidates.

For each split and query, fixed candidate pools contain all role-specific recorded positives and up to 100 unrecorded targets sampled from that split's target universe using random, length, terminal-motif or edit-neighbor matching. Full-target retrieval uses the same universe. Other recorded partners are excluded from distractors without being added as role-specific test positives. All scorers and model seeds share frozen candidate lists. Unlike free generation, these retrieval pools include the true targets by construction. Score ties are resolved by lexicographic target-sequence order. In the primary chain-disjoint tests, every training-frequency score is zero, so that baseline is a deterministic tie-order control, not informative frequency prediction.

Direction-specific score means and population standard deviations are estimated on tuning candidates. A ranker fusion weight from $\{0, .25, .5, .75, 1\}$ is shared across directions and selected by direction-balanced query-macro Hit@5, with ties favoring less ranker weight. Free and fixed-pool calibration are separate. An independent calibration subset selects the acceptance threshold by balanced accuracy against recorded positives and constructed unrecorded combinations; ties favor the highest threshold, including reject-all. No-positive or no-negative calibration yields reject-all. Acceptance curves apply fixed calibration-score quantiles and both endpoints to test data. Pooled AUROC and AUPRC are reported with positive prevalence, consistent with the limitations of imbalanced classification summaries (see the precision–recall reference in the main manuscript). These labels and scores do not define biological specificity or binding probability.

S6 Metric definitions, uncertainty and distance strata

For query q, let P_q be its recorded test partners, C_q its unique valid candidates and $C_q^{(K)}$ its first K ranked candidates. Define

$$\text{Coverage}(q) = \mathbf{1}(P_q \cap C_q \neq \varnothing), \qquad H_K(q) = \mathbf{1}(P_q \cap C_q^{(K)} \neq \varnothing). \tag{3}$$

$$R_K(q) = \frac{|P_q \cap C_q^{(K)}|}{|P_q|}. \tag{4}$$

If $r_q(t)$ is the one-based candidate rank, reciprocal rank is $1/\min_{t \in P_q \cap C_q} r_q(t)$ when a partner is present and zero otherwise. Mean positive rank is $|P_q|^{-1} \sum_{t \in P_q} \widetilde{r}_q(t)$, where $\widetilde{r}_q(t) = r_q(t)$ for present partners and $|C_q| + 1$ for absent partners. MRR averages reciprocal rank across queries. We report query-macro outcomes and positive-count-weighted query outcomes; the latter yields edge-micro Recall. The primary endpoint is the equal-direction mean of query-macro Hit@5 at 64 draws, reported separately for likelihood, ranker and fused ordering. Per-fit event counts and query-level results appear in Additional file 4; all frozen retrieval pools, candidate hashes and split assignments are supplied in Additional file 5. Ranker and fused ordering assess the same generator candidates and do not constitute additional generation methods. NLL, token-weighted NLL, raw ordering, validity, uniqueness, novelty and diversity are secondary summaries.

The reported query-macro NLL first averages the target-token losses for each recorded partner, including EOS and excluding padding and the chain-control prediction. For target sequence y, let $T_y = |y| + 1$, with EOS at position T_y. For the query set Q_d in direction d, the implemented hierarchy is

$$\ell(q, y) = -\frac{1}{T_y} \sum_{i=1}^{T_y} \log p_\theta(y_i \mid q, y_{<i}), \tag{5}$$

$$L_d = \frac{1}{|Q_d|} \sum_{q \in Q_d} \frac{1}{|P_q|} \sum_{y \in P_q} \ell(q, y), \qquad L_{\text{macro}} = \frac{L_{\gamma \to \delta} + L_{\delta \to \gamma}}{2}. \tag{6}$$

Thus, partners within a query and queries within a direction receive equal weight; longer targets do not receive more weight in this summary. Checkpoint selection uses this same hierarchy on the selection partition (one direction only for directional fits). The separately reported token-weighted NLL instead divides the sum of all recorded target-token losses by their total token count within each direction. The optimization loss also averages non-ignored target tokens within each training minibatch; it is distinct from the query-macro reporting hierarchy.

We use 2000 paired cluster-bootstrap replicates over complete test components, or source-query groups for directional tasks. Joint directional endpoints use the same component resample but average direction-specific query means equally. Model comparisons align query identities, positives and cluster labels; fixed pools must also have identical hashes. Two model seeds are averaged within each split before the descriptive mean and sample standard deviation across the three split means are calculated. These SDs are not confidence intervals. Repeated splits and seeds are not independent biological subjects. Intervals condition on fitted models and saved candidates, and are descriptive without family-wise multiplicity adjustment. Zero-event percentile intervals of $[0, 0]$ are not population upper bounds; an explicitly labeled iid Wilson reference is provided for binary query events only, not fractional Recall.

Every paired chain is compared with supervised and pretraining references using raw and length-normalized Levenshtein distances on complete CDR3, internal residues (excluding three residues at each end) and terminal residues. Exact minimum witnesses are saved. Full-sequence strata are 0, 1, 2 and at least 3. The distant subset requires both source and eligible recorded target to be at distance at least three from both reference pools. Edge-defined strata retain the

original candidate lists and test-cluster identities. Empty and single-cluster strata are explicit. Source, length, number of partners and training-target overlap are additional descriptive strata.

S7 Pretraining and negative-sampling sensitivity

S7.1 Pretraining mixture and downstream sampling have distinct effects

The two-seed downstream NLL means for square-root, raw and balanced pretraining are, respectively, 1.256/0.754, 1.255/0.757 and 1.259/0.755 for $\gamma \to \delta / \delta \to \gamma$ on split 17 (Figure S1). Raw-minus-square-root equal-direction differences are approximately 0.001 nats per token in both fits, with paired intervals crossing zero. Balanced-minus-square-root differences are approximately 0.002 in both fits, with intervals above zero. These small NLL differences do not select a new primary mixture after testing.

Restricting pretraining to one chain has larger directional effects. Gamma-only initialization gives downstream NLL 1.294/0.755, while delta-only initialization gives 1.257/0.779. On frozen denoising validation, gamma-only pretraining yields 39.6% selected-site accuracy for delta, and delta-only pretraining yields 27.3% for gamma, compared with 64.4% and 70.8% under the square-root mixture. The results are consistent with chain-specific reconstruction exposure affecting the target-sequence prior. Equal total sequence draws do not equate chain-specific or token exposure, and these teacher-forced differences do not demonstrate reliable free partner recovery.

Edge-uniform paired training gives NLL 1.263/0.759, compared with 1.256/0.754 under query-uniform training. Its equal-direction excess is approximately 0.006 nats per token in both fitted seeds, with paired intervals above zero. The nominally direction-budget-matched unidirectional models give NLL 1.258 for $\gamma \to \delta$ and 0.756 for $\delta \to \gamma$, close to the bidirectional model. All these settings retain sparse free-recovery events. Their actual early-stop and token exposure records accompany the matched nominal budgets; the comparison does not establish a universal benefit from sharing the two directions.

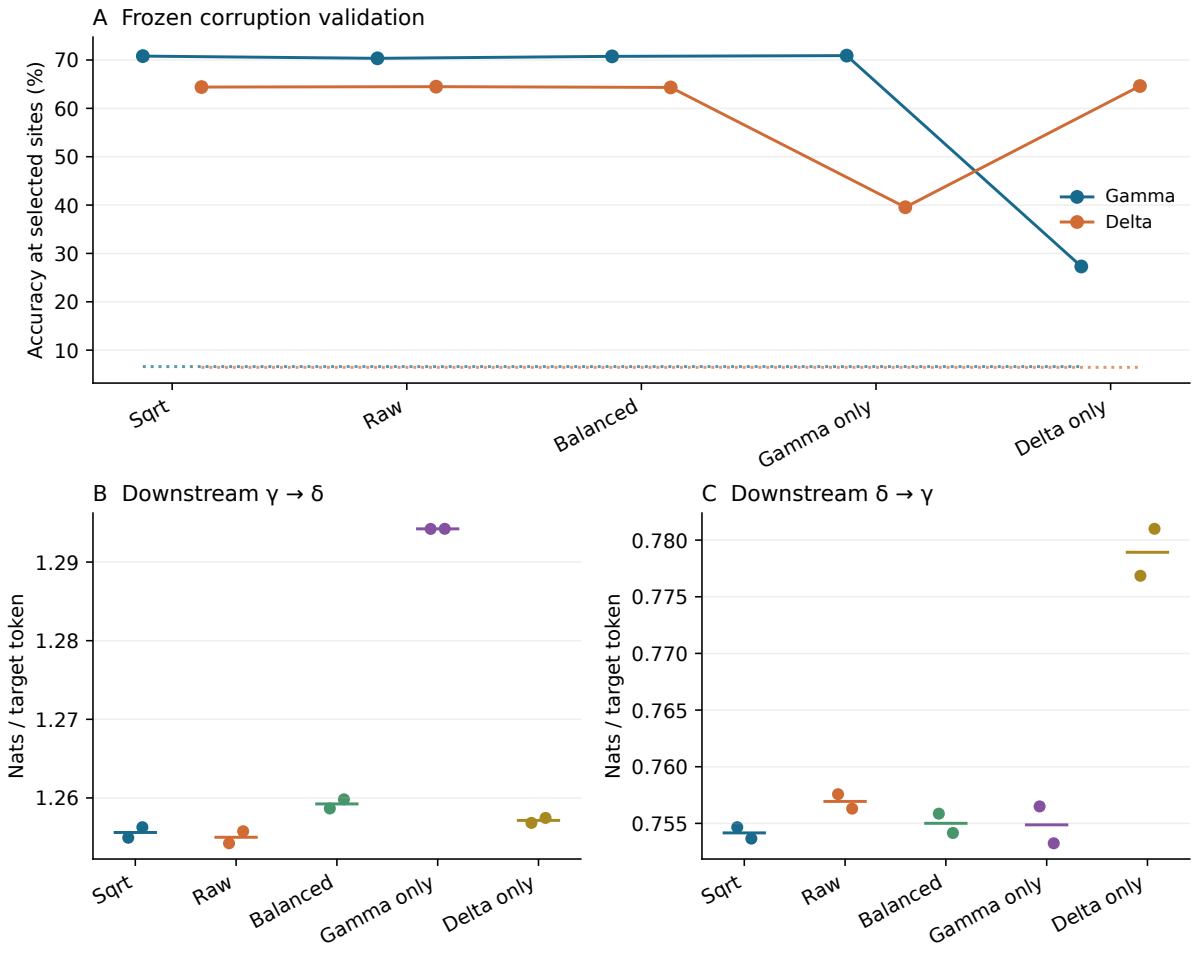


Figure S1: Pretraining mixture sensitivity. Dots retain fitted seeds. Equal sequence draws do not imply equal chain-specific or token exposure. Reconstruction is a single-chain endpoint.

S7.2 Changing negative training does not resolve candidate scarcity

Random-only and mixed negative training at ratios one, three and five reuse the same split-17 generator candidates and retrieval pools (Figure S2). All twelve fitted rankers have zero free-generation $\gamma \to \delta$ Hit@5. For $\delta \to \gamma$, individual fitted values range from zero to 0.22% (zero to two of 900 queries), with no monotonic gain from increasing the ratio. The primary mixed/3 setting averages 0.11%; random/5, mixed/1 and mixed/5 each average 0.17%. Such small event differences do not establish a reproducible recovery advantage: paired intervals for every alternative negative setting versus the primary setting include zero in both directions and both fitted seeds.

Fixed-pool differences are larger but depend on the comparison. In random retrieval pools, random/3 gives two-seed mean ranker Hit@5 of 10.8% for $\gamma \to \delta$, compared with 8.2% under mixed/3. For $\delta \to \gamma$, random/5 gives 9.1%, compared with 7.1% under mixed/3. These descriptive rankings are not uniform across pools or seeds, and no setting supplies candidates absent from the frozen free-generation set. Negative-draw and fallback counts, selected steps and paired contrasts are retained in the numerical supplement. None of these constructed-negative comparisons identifies biological incompatibility.

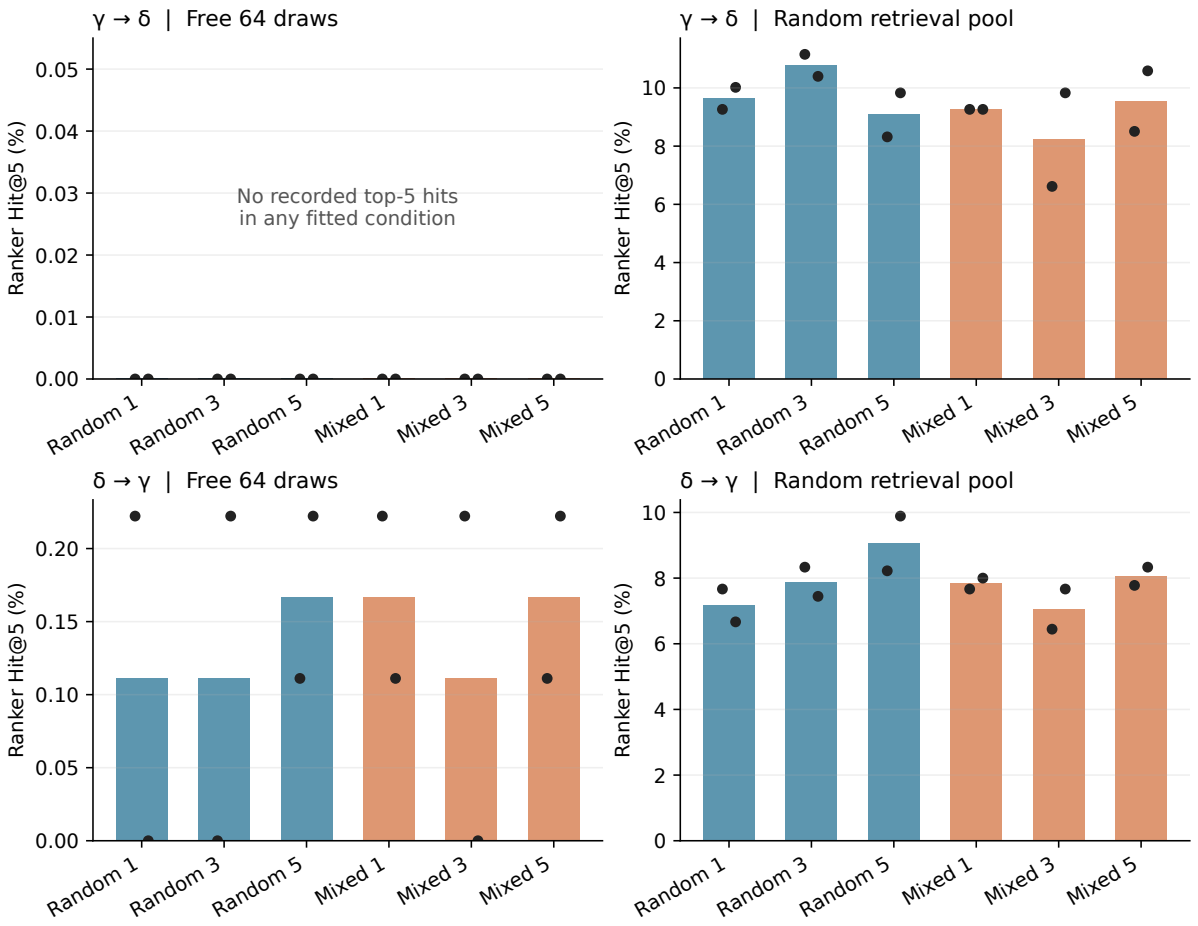


Figure S2: Ranker negative-sampling sensitivity. All settings reuse the same saved free candidates and frozen truth-containing retrieval pools. No ranker adds absent candidate sequences.

S8 Nearest-input baselines and exposure diagnostics

Nearest-input transfer uses the normalized full-CDR3 distance, lexicographic input ties and training-degree/lexicographic target ordering. One nearest input is primary; five is a fixed sensitivity. At most 64 unique training targets are returned. In seen-input/withheld-edge tasks the exact input is its own nearest neighbor, so one-neighbor transfer cannot recover its withheld edges. Primary component-disjoint transfer is also structurally zero because all test targets are absent from training. These zero outcomes do not identify a learned-model mechanism. The separate nearest-input partner-similarity retrieval score extends to the role-specific target universe using minimum target edit distance to the nearest input's known training partners. Neither neighborhood size nor distance rule was selected using test outcomes. Test memberships, query-balanced selection and the giant-component edge reservation are frozen in the revision split manifest. These are exploratory diagnostics specified after the primary analysis.

S8.1 Unseen-gamma validation design and historical estimates

The original selection partition contained one gamma query with 581 recorded partners and no ranking distractors. Its ranker selection MRR was identically one. Original conditional likelihood Hit@5 was 9/64 in each seed (14.06%), compared with 9/64 and 8/64 for constant source (14.06%/12.50%). These estimates are retained for traceability, not as robust unseen-gamma evidence. The query-balanced-selection refits preserve the original train, aggregate validation and test memberships and replace this diagnostic in the main text. Primary component-disjoint results remain unchanged. The query-balanced selection has 31 gamma queries, 275 edges, a largest query with 137 partners and at least 118 eligible distractors per query. The tuning and calibration roles each contain 16 disjoint gamma queries and are not used for new tuning for these refits.

S8.2 Source and component-stratified descriptive analyses

Additional file 2 includes primary source and source-distance strata and a post-hoc source-balanced description. The latter averages nonempty source membership strata equally. Memberships can overlap, and very small source strata are unstable; this is neither a reweighted primary endpoint nor a source-independent inferential comparison. Post-hoc component-size tables group primary-test components into 1, 2–5, 6–20 and more than 20 edges, retaining the original query positives and saved candidates. Nonempty bins give counts and descriptive means without inferential intervals. The distant-edge recheck recovers 5, 5 and 3 primary edges, with no new claim of distant-sequence transfer. Within-giant-component estimates originate from one original component and deliberately have no inferential interval. Additional file 4 retains every diagnostic query outcome, including zero events, all control conditions and paired contrasts. Additional file 3 records actual generator sequences and target tokens; ranker encoded-token counts were not logged and are not imputed.

S8.3 Query-balanced unseen-gamma evaluation

The query-balanced validation has 31 selection queries with eligible distractors, and ranker selection MRR is no longer identically one: the selected values are 0.1405 and 0.1618. Training and test memberships remain unchanged. At 64 free draws, conditional likelihood recovers 8/64 and 10/64 queries, compared with 8/64 and 9/64 for constant source, 10/64 and 8/64 for random initialization, and 6/64 and 7/64 for shuffled input. These counts also equal coverage for each fitted generator. Conditional-minus-prior Hit@5 differences are 0.00 percentage points (95% source-query interval $[0.00, 0.00]$) and 1.56 points ($[0.00, 4.69]$). The small number of discordant events limits inference. Ranker ordering recovers 6/64 and 8/64 queries, two fewer than likelihood in each seed. One-neighbor and five-neighbor direct partner transfer recover 3/64 and 7/64 queries, respectively. The repaired test still permits target exposure: 252 distinct recorded target deltas also occur in training. These query-balanced results supersede the original single-query-selection diagnostic, whose estimates remain in Section S8.1. They do not establish a reproducible conditional free-recovery advantage over the matched prior.

S8.4 Within-component withheld-edge evaluation

Within the exposed giant component, conditional coverage is 70/177 and 62/177 gamma queries and 155/261 and 164/261 delta queries. Likelihood Hit@5 is 60/177 and 50/177 for $\gamma \to \delta$ (33.90%/28.25%), and 123/261 and 118/261 for $\delta \to \gamma$ (47.13%/45.21%; main Figure 6). Constant-source Hit@5 is 47/177 in both gamma-query fits and 113/261 and 121/261 for delta queries. Shuffled-input counts are 57/177 and 54/177, and 126/261 and 116/261, respectively. Random initialization yields 32/177 and 27/177, and 121/261 and 124/261. Thus, much of the high recovery persists without the correct input association; a conditional-minus-prior advantage is not consistent across directions and seeds.

Ranker ordering gives 30/177 and 35/177 gamma-query hits and 71/261 and 94/261 delta-query hits, below likelihood in all four fits. Direct transfer from one nearest input has zero hits in both directions: the nearest input is the query itself, whose remaining training partners exclude its withheld edges. The fixed five-neighbor sensitivity recovers 4/177 and 57/261 queries. These proposals contain copied training partners and use a different candidate mechanism from stochastic generation. All 500 test edges retain both endpoints in supervised training and originate from one original graph component. The results are point estimates and counts, without an inferential component-bootstrap interval. Their contrast with the primary task describes different graph populations and training composition, rather than distant or unseen-component transfer.

S8.5 Directional exposure comparisons

Other directional tasks also have higher recovery than the both-chain-unseen primary task (main Figure 6). For unseen delta inputs, conditional $\delta \to \gamma$ Hit@5 is 39.75%/38.51%, compared with 41.61%/39.13% for the prior. This test has 161 queries and 244 distinct recorded target gammas shared with training. Thus, substantial recovery in a target-exposed task does not itself demonstrate an input-specific advantage. In seen-input/unseen-edge tests, conditional Hit@5 is 6.89%/6.67% for $\gamma \to \delta$ versus 5.33%/5.33% for the prior, and 32.58%/34.09% for $\delta \to \gamma$

versus 30.30%/29.92%. These populations, edge counts and permissible overlaps differ from the primary test and are reported as distinct diagnostics, with source-query-group intervals. For seen delta inputs, conditional-minus-prior Hit@5 differences are 2.27 percentage points (95% interval $[-1.52, 6.06]$) and 4.17 points ($[0.38, 8.33]$). The latter fitted contrast is positive, but the two-seed result is not uniform and does not transfer to both-chain-unseen generation.

S8.6 Study-level exposure comparisons

Strict study holdout has 207 gamma and 419 delta queries over 626 pairs. Conditional $\delta \to \gamma$ likelihood Hit@5 is 11.93%/10.26%, compared with 6.68%/6.44% for constant source and 8.83%/9.79% for shuffled input. Conditional-minus-prior differences are 5.25 percentage points with a 95% component interval $[0.00, 6.23]$ and 3.82 points with $[0.00, 4.48]$; both intervals reach zero despite positive fitted differences. When shared supervised chains are retained, the corresponding conditional values increase to 47.49%/44.39%; constant source already reaches 43.68%/43.20%, and shuffled input reaches 43.68%/43.91%. The shared-source conditional-minus-prior differences are 3.82 percentage points ($[-2.13, 4.69]$) and 1.19 points ($[0.00, 1.42]$), again with paired component intervals including zero. For $\gamma \to \delta$, conditional likelihood Hit@5 is zero in both seeds under both source protocols. Source results therefore cannot be reduced to uniform failure, nor can high $\delta \to \gamma$ recovery be attributed entirely to conditioning.

The source test contains 90 components, with 529 of 626 edges and 323 of 419 delta queries in its largest component. A post-hoc description of the saved query outcomes finds 50 of 50 and 42 of 43 strict-source conditional top-five hits in that component. The corresponding shared-source counts are 198 of 199 and 184 of 186. This concentration motivates the clustered uncertainty rather than treating all hits as independent replications. The source protocols share the same test set and filtered pretraining pool, but supervised training contains 4956 versus 7597 edges. Shared-source training overlaps the test in 144 gamma and 13 delta sequences. Exposure and training composition therefore change together. Neither protocol supports a donor-independent or distant-sequence generalization claim.

Ranker-only $\delta \to \gamma$ Hit@5 is 3.10%/2.86% under strict source holdout and 14.32%/13.60% under shared source, below likelihood in all four fits. The higher source-task recovery consequently does not support a separate-ranking benefit.

S9 Sampling budgets, decoding and calibration

Split-17 conditional $\delta \to \gamma$ coverage increases from 0.67%/0.33% at 16 draws to 2.33%/1.44% at 64 and 6.33%/6.22% at 256 for model seeds 3407/3508. Nevertheless, with training targets allowed, likelihood Hit@5 is 0.22%/0.11%, 0.22%/0.00% and 0.00%/0.00%, respectively (Figure S3). Additional samples can introduce high-scoring competitors as well as recorded partners. The 256 draws average approximately 203 unique valid gamma candidates per query, but only about 53% of raw gamma draws differ from supervised-training targets. For $\gamma \to \delta$, approximately 88% of draws are novel and about 244 unique candidates remain at 256 draws; coverage is 0.57% in each seed and likelihood Hit@5 remains zero.

Excluding exact training targets changes ordering without increasing the sampling budget.

Because primary-test targets are chain-disjoint from training, their coverage is unchanged. At 256 draws, $\delta \to \gamma$ likelihood Hit@5 increases to 1.89%/2.44%, while $\gamma \to \delta$ reaches 0.57%/0.19%. This policy result is reported separately from the frozen primary endpoint. It cannot be applied unchanged as a general advantage in shared-target tasks, where exclusion may remove a legitimate test partner.

The six-cell tuning-only temperature/top-p grid has zero likelihood Hit@5 in both directions for both fitted seeds at 64 draws. The prespecified tie rule retains temperature one and top-$p = 0.95$; these sparse tuning outcomes do not establish an improved decoding setting. All grid cells and their coverage, validity, uniqueness and novelty are included in Additional file 4.

Free-candidate calibration contains no recorded $\gamma \to \delta$ positive in any primary fit, triggering the specified reject-all rule. The $\delta \to \gamma$ calibration sets contain only one to six recorded positive candidates. Their selected likelihood/fused thresholds accept at least one candidate for every test query, without improving the very sparse top-five hits. These calibration data do not establish a useful operating point for biological candidate selection.

Deterministically chosen descriptive examples illustrate both failure stages. For split 17/model seed 3407 in $\delta \to \gamma$, 879 of 900 queries lack a recorded partner in the 64-draw candidate set, 19 contain one outside the likelihood top five, and two yield a top-five hit. In $\gamma \to \delta$, the corresponding counts are 527, two and zero out of 529 queries. Additional file 4 records the first input in lexicographic order within each nonempty category and preserves its positives, ranked candidates and training-target flags. These examples are post-hoc descriptions, not additional independent validation.

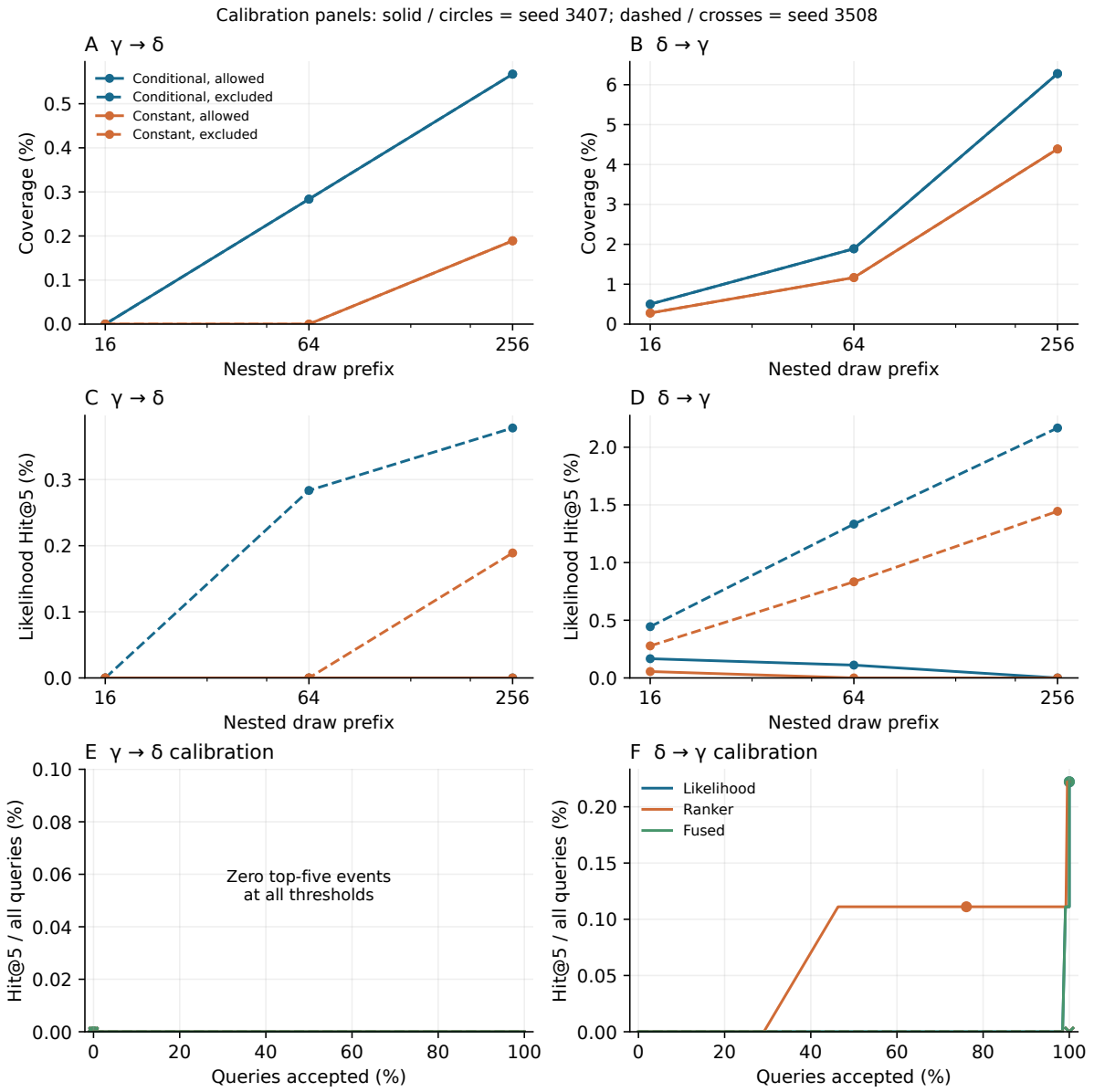


Figure S3: Sampling budgets and exploratory calibration. The 16/64/256 budgets are prefixes of the same saved draws on split 17. The first two rows separate candidate coverage from likelihood Hit@5, with exact training targets allowed (solid) or excluded (dashed); lines average two seeds. Coverage is unchanged by target exclusion in the chain-disjoint primary test. The last row shows exploratory calibration-quantile operating curves; solid/dashed lines identify seeds and markers identify independently calibrated thresholds. All test queries remain in recovery denominators. More draws can increase coverage without improving top-five ordering. No threshold is a validated biological selector.